

CS492D: Diffusion Models and Their Applications

Denoising Diffusion Implicit Models

LECTURE 5
MINHYUK SUNG

Fall 2024
KAIST

Quiz Policy

- For math questions, please try to solve the problem using pen and paper (or a tablet). Do not copy formulations from any sources.
- A time limit will be set for each quiz.
- It is acceptable to post an incorrect answer; just share your derivation within the time limit.
- Your participation will be assessed based on whether you posted an answer, not on its correctness.
- If you are unable to solve the problem within the time limit, it is strongly recommended that you try to solve it after the lecture.

Previously in CS492D

Variational Diffusion Models

1. Both forward (encoding) and reverse (decoding) processes are **sequential** processes (with a sequence of latent variables).
2. The **forward process** is not learned but predefined.
3. The **dimensions** of the latent variables and the input data are the **same**.

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Denoising Diffusion Probabilistic Models (DDPM)

1. The sequential forward and reverse processes are **Markovian** processes:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0)$$

2. The **forward transitional distribution** is defined as follows:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

where $\{\beta_t \in (0, 1)\}_{t=1}^T$ and $\beta_1 \leq \beta_2 \leq \dots \leq \beta_T$. $\beta_t \ll 1$.

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Denoising Diffusion Probabilistic Models (DDPM)

1. $q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I})$
where $\alpha_t = 1 - \beta_t$.
2. $q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$
where $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$.
3. $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$
where $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0$ and $\tilde{\sigma}_t^2 = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$.

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$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$$

$$1. \quad \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} \mathbf{x}_0 \text{ or}$$

$$2. \quad \tilde{\mu}(\mathbf{x}_t, \boldsymbol{\epsilon}_t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_t \right)$$

$$\text{where } \boldsymbol{\epsilon}_t = \frac{1}{\sqrt{1-\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0)$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t} \boldsymbol{\epsilon}_t$$

Previously in CS492D

Variational Distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$

1. Mean predictor $\mu_{\theta}(\mathbf{x}_t, t)$:

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_{\theta}(\mathbf{x}_t, t), \tilde{\sigma}_t^2 \mathbf{I})$$

2. $\mathbf{x}_0$ predictor $\hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t)$

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t), \tilde{\sigma}_t^2 \mathbf{I}\right)$$

3. $\boldsymbol{\epsilon}_t$ predictor $\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)\right), \tilde{\sigma}_t^2 \mathbf{I}\right)$$

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Loss Functions

1. Mean predictor $\mu_\theta(\mathbf{x}_t, t)$:

$$\mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), t > 1, \mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_0)} [\|\mu_\theta(\mathbf{x}_t, t) - \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)\|^2]$$

2. $\mathbf{x}_0$ predictor $\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)$

$$\mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), t > 1, \mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_0)} [\|\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t) - \mathbf{x}_0\|^2]$$

3. $\boldsymbol{\varepsilon}_t$ predictor $\hat{\boldsymbol{\varepsilon}}_\theta(\mathbf{x}_t, t)$

$$\mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), t > 1, \mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_0)} [\|\hat{\boldsymbol{\varepsilon}}_\theta(\mathbf{x}_t, t) - \boldsymbol{\varepsilon}_t\|^2]$$

Previously in CS492D

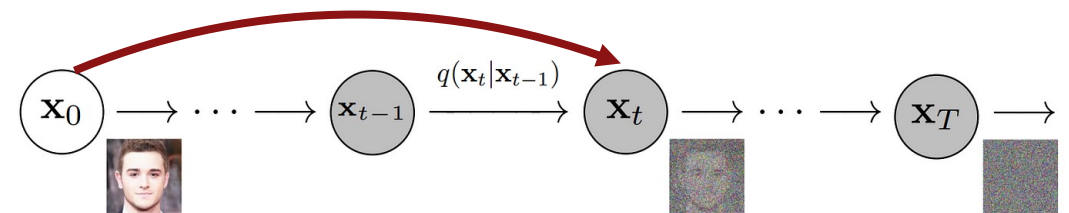
Training with ϵ_t Predictor

Repeat:

1. Take a random $\mathbf{x}_0$.
2. Sample $t \sim \mathcal{U}(\{1, \dots, T\})$.
3. Sample $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. Compute $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t$.
5. Take gradient descent step on $\nabla_{\theta} \|\hat{\epsilon}_{\theta}(\mathbf{x}_t, t) - \epsilon_t\|^2$.

Same as sampling

$$\mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$



Previously in CS492D

Reverse Process with ε_t Predictor

1. Sample $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
2. For $t = T, \dots, 1$, repeat:

1. Compute $\tilde{\boldsymbol{\mu}} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \hat{\boldsymbol{\varepsilon}}_{\theta}(\mathbf{x}_t, t) \right)$.
2. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
3. Compute $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \tilde{\sigma} \mathbf{z}_t$.

Same as sampling
 $\mathbf{x}_{t-1} \sim p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t)$.

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Q. Given $\mathbf{x}_t$ and the noise prediction $\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$, what is the prediction of $\mathbf{x}_0$ (which will be denoted as $\mathbf{x}_{0|t}$)?

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$$\mathbf{A.} \mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) \right)$$

Previously in CS492D

Reverse Process with ε_t Predictor

1. Sample $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
2. For $t = T, \dots, 1$, repeat:
 1. Compute $\mathbf{x}_{0|t}$ from $\mathbf{x}_t$ and $\hat{\varepsilon}_\theta(\mathbf{x}_t, t)$.
 2. Compute $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_{0|t})$.
 3. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
 4. Compute $\mathbf{x}_{t-1} = \tilde{\mu} + \tilde{\sigma} \mathbf{z}_t$.

Same as sampling
 $\mathbf{x}_t \sim p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)$.

Direct Sampling vs. Generative Models

Q. Why don't we directly sample from the training data instead of training a generative model?



Direct Sampling vs. Generative Models

- **Efficiency and Scalability:**

Generative models can efficiently encode the data distribution into a neural network, eliminating the need to store the entire dataset.

- **Novelty and Diversity / Privacy and Security Concerns:**

Generative models can generate new samples that are not identical to any specific training data but still appear plausible.

- **Adaptability to New Conditions:**

For conditional generation, generative models can be more efficient at generating data based on the given conditions compared to retrieval methods.

Today's Topics

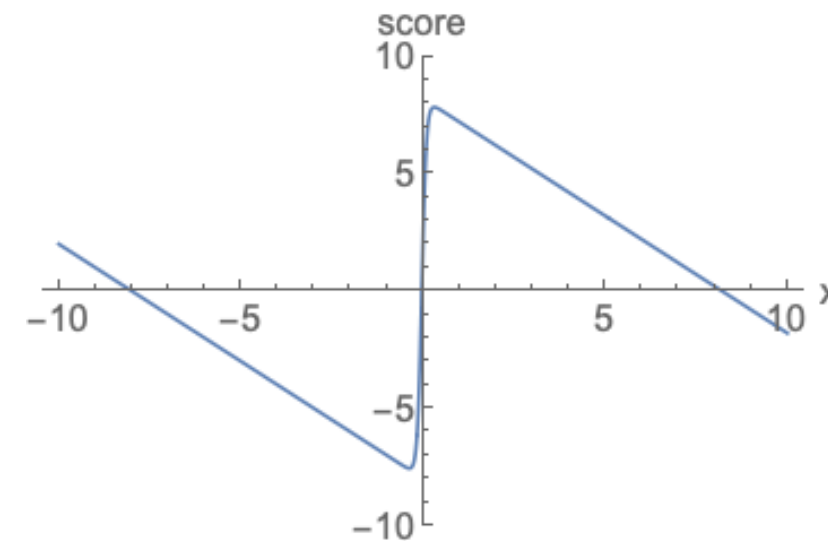
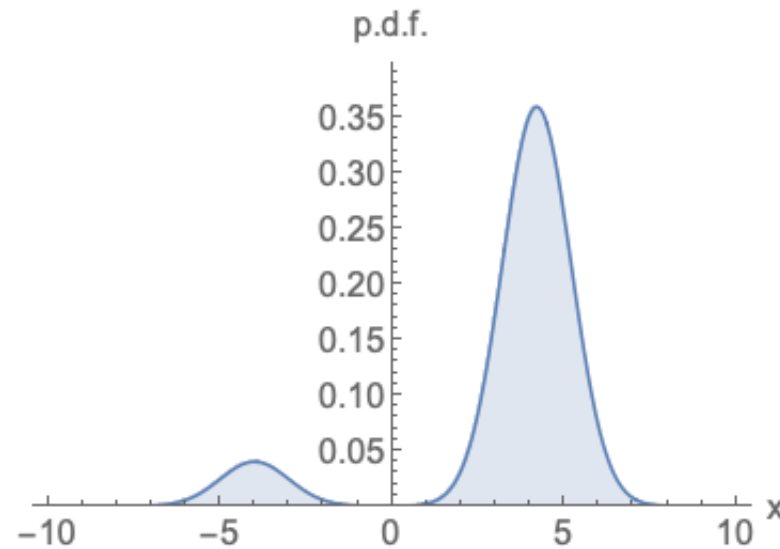
- Connection to Score-Based Models
- Denoising Diffusion Implicit Models (DDIM)

Connection to Score-Based Models

(Stein) Score Function

The (Stein) score (in statistics) is the **gradient of the log-likelihood** function with respect to a data point:

$$\nabla_x \log p(x)$$



(Stein) Score Function

- What is the **score** of $q(\mathbf{x}_t|\mathbf{x}_0)$: $\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0)$?
- How is it related to $\boldsymbol{\epsilon}_t$?

(Stein) Score Function

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$$

The **score** of $q(\mathbf{x}_t|\mathbf{x}_0)$:

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) = \nabla_{\mathbf{x}} \left(-\frac{\|\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0\|^2}{2(1 - \bar{\alpha}_t)} \right) = -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{1 - \bar{\alpha}_t}$$

(Stein) Score Function

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$$

$$\boldsymbol{\varepsilon}_t = \frac{1}{\sqrt{1 - \bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0)$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{1 - \bar{\alpha}_t} = -\frac{\boldsymbol{\varepsilon}_t}{\sqrt{1 - \bar{\alpha}_t}}$$

(Stein) Score Function

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) = -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0}{1 - \bar{\alpha}_t} = -\frac{\boldsymbol{\epsilon}_t}{\sqrt{1 - \bar{\alpha}_t}}$$

*The **noise predictor** $\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$ can be interpreted as predicting the score $\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0)$, up to a scaling factor.*

Tweedie's Formula

How to estimate the **true mean** of a normal distribution from samples drawn from it?

For $\mu \sim p(\mu)$ and $x|\mu \sim \mathcal{N}(x; \mu, \Sigma)$,

$$\mathbb{E}[\mu|x] = x + \Sigma \nabla_x \log p(x)$$

Tweedie's Formula

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

Given the Tweedie's formula,

$$\mathbb{E}[\boldsymbol{\mu} | \mathbf{x}_t] = \sqrt{\bar{\alpha}_t} \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t] = \mathbf{x}_t + (1 - \bar{\alpha}_t) \nabla_{\mathbf{x}} \log q(\mathbf{x}_t)$$

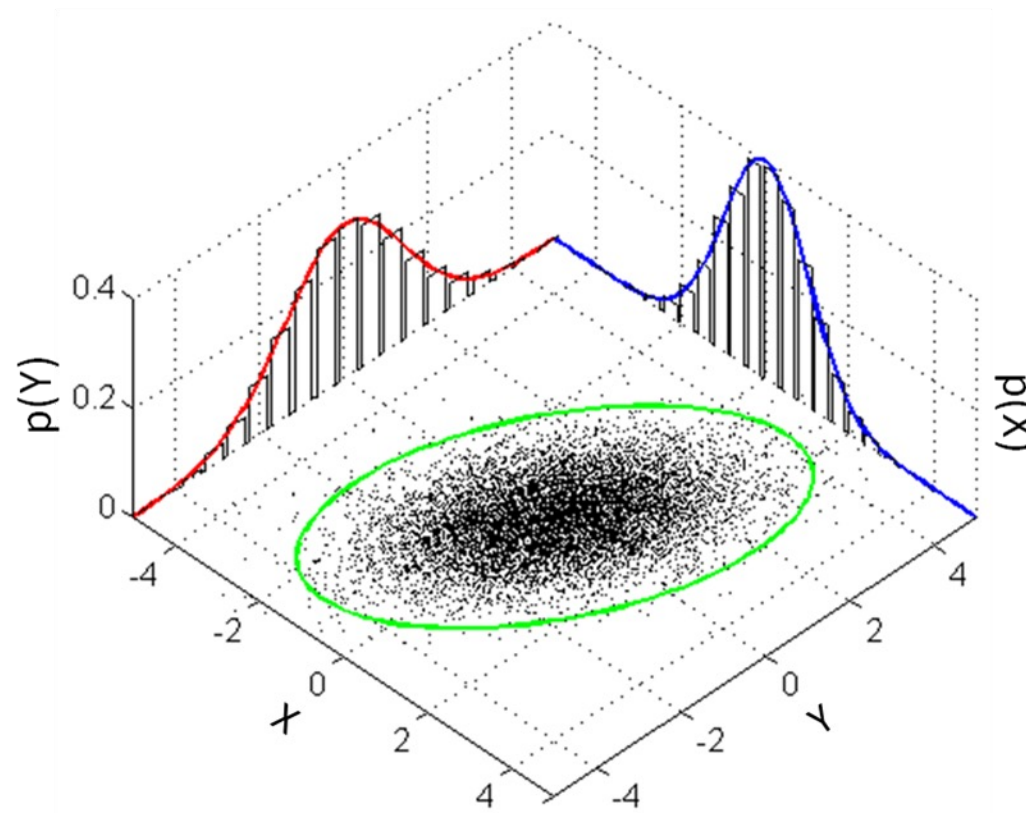
$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t) = - \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t]}{1 - \bar{\alpha}_t}$$

Langevin Dynamics

Why is the **score** function important?

Recall Lecture 2

If the **probability density function (PDF)** of the **distribution** is known, we can sample from it directly.



Langevin Dynamics

Why is the **score** function important?

Even without knowing $q(\mathbf{x})$, if we have the score function $\nabla_{\mathbf{x}} \log q(\mathbf{x})$, we can sample from the distribution $q(\mathbf{x})$ using **Langevin dynamics**!

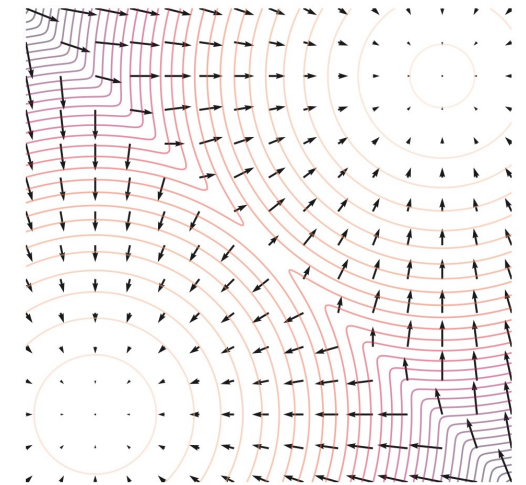
Langevin Dynamics

*Similar to the
Reverse process network!*

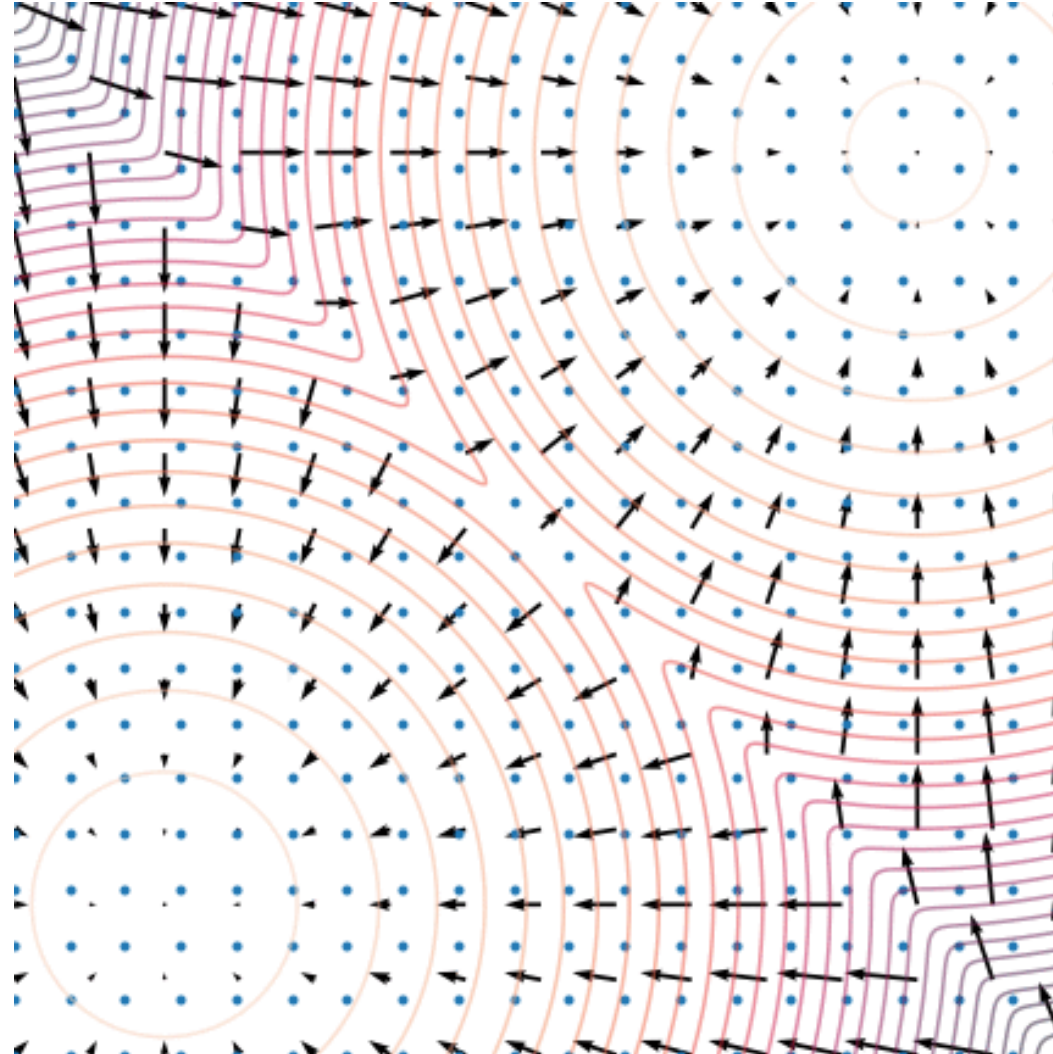
- Sample $\mathbf{x}$ from a prior distribution.
- Iterate the following procedure T steps:

$$\mathbf{x} \leftarrow \mathbf{x} + \eta \nabla_{\mathbf{x}} \log q(\mathbf{x}) + \sqrt{2\eta} \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

- It converges to $q(\mathbf{x})$ when $\eta \rightarrow 0$ and $T \rightarrow \infty$.
- Note that it only uses the **score** information $\nabla_{\mathbf{x}} \log q(\mathbf{x})$, not $q(\mathbf{x})$.



Langevin Dynamics



Score Matching

*Similar to the
noise prediction network!*

One possible way to encode the data distribution $q(\mathbf{x})$ into a neural network is to train a **score prediction** network $s_\theta(\mathbf{x})$ using the following loss function:

$$\mathbb{E}_{\mathbf{x} \sim q(\mathbf{x})} [\|s_\theta(\mathbf{x}) - \nabla_{\mathbf{x}} \log q(\mathbf{x})\|^2]$$

Noise-Conditional Score-Based Models

How do we compute $\nabla_{\mathbf{x}} \log q(\mathbf{x})$ when we only have samples of $q(\mathbf{x})$?

Now let $q(\mathbf{x}) = q(\mathbf{x}_0)$.

Approximate $q(\mathbf{x}_0)$ as a **mixture of Gaussians**:

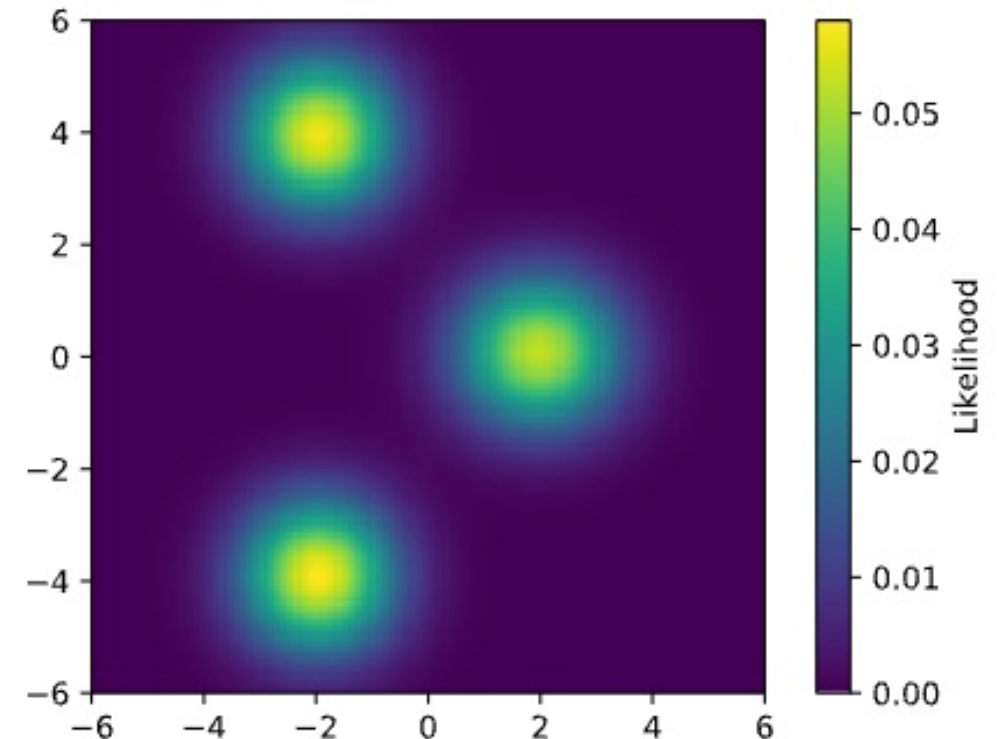
$$q(\mathbf{x}_t) = \int q(\mathbf{x}_0) q(\mathbf{x}_t | \mathbf{x}_0) d\mathbf{x}$$

where $q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \mathbf{x}_0, \sigma_t^2 \mathbf{I})$.

Noise-Conditional Score-Based Models

$$q(\mathbf{x}_t) = \int q(\mathbf{x}_0)q(\mathbf{x}_t|\mathbf{x}_0)d\mathbf{x}$$

where $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \mathbf{x}_0, \sigma_t^2 \mathbf{I})$.



Noise-Conditional Score-Based Models

$$q(\mathbf{x}_t) = \int q(\mathbf{x}_0)q(\mathbf{x}_t|\mathbf{x}_0)d\mathbf{x}$$

Sampling from $q(\mathbf{x}_t)$ is the same as

1. sampling from $q(\mathbf{x}_0)$ (taking a random $\mathbf{x}_0$) and then
2. sampling from $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \mathbf{x}_0, \sigma_t^2 \mathbf{I})$.

Noise-Conditional Score-Based Models

$$\mathbb{E}_{\mathbf{x}_t \sim q(\mathbf{x}_t)} \left[\left\| s_{\theta}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t) \right\|^2 \right]$$

$$= \dots$$

$$= \mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), \mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_0)} \left[\left\| s_{\theta}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|^2 \right]$$

Identical to the loss function of DDPM, up to scale!

Noise-Conditional Score-Based Models

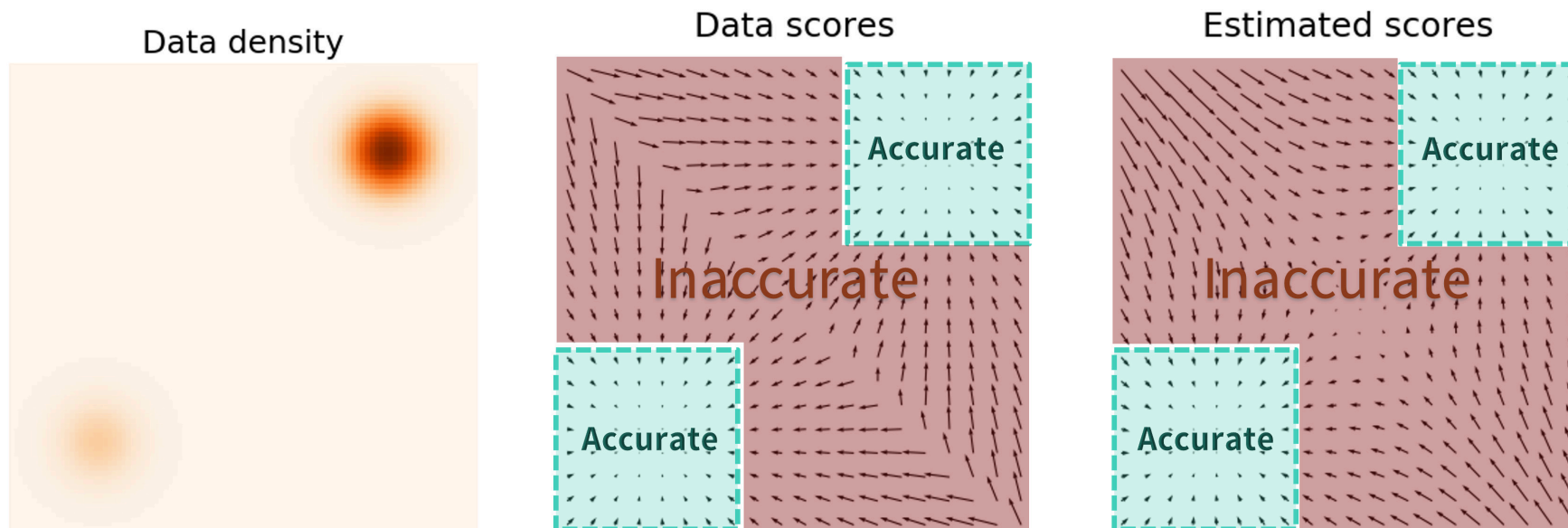
$$q(\mathbf{x}_t) = \int q(\mathbf{x}_0) \mathcal{N}(\mathbf{x}_t; \mathbf{x}_0, \sigma_t^2 \mathbf{I}) d\mathbf{x}$$

As σ_t is **small**,

- **(+)** Close to the given data samples.
- **(−)** May lead to low-density regions where the score prediction could be inaccurate.

Noise-Conditional Score-Based Models

$$\mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0)} \left[\left\| \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_0) - s_{\theta}(\mathbf{x}_t) \right\|^2 \right]$$



Noise-Conditional Score-Based Models

$$q(\mathbf{x}_t) = \int q(\mathbf{x}_0) \mathcal{N}(\mathbf{x}_t; \mathbf{x}_0, \sigma_t^2 \mathbf{I}) d\mathbf{x}$$

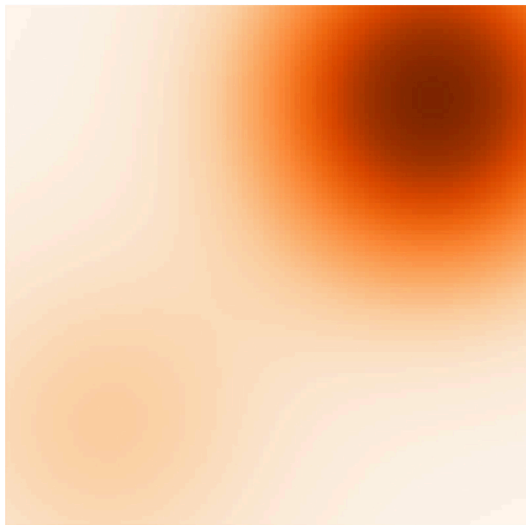
As σ_t is **large**,

- **(+)** Can help avoid low-density regions.
- **(−)** May over-corrupt the original data distribution, also leading to noisy score predictions.

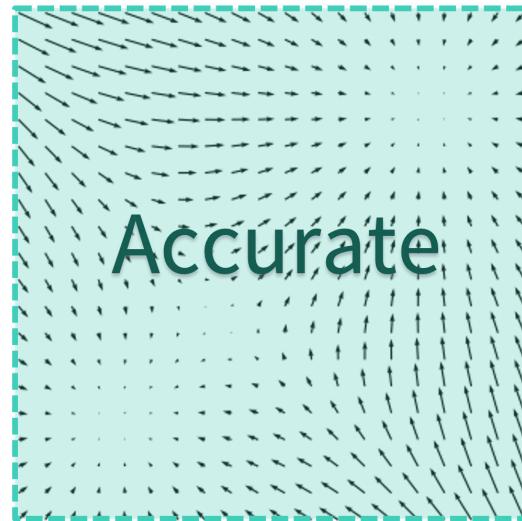
Noise-Conditional Score-Based Models

$$\mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0)} \left[\left\| \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_0) - s_\theta(\mathbf{x}_t) \right\|^2 \right]$$

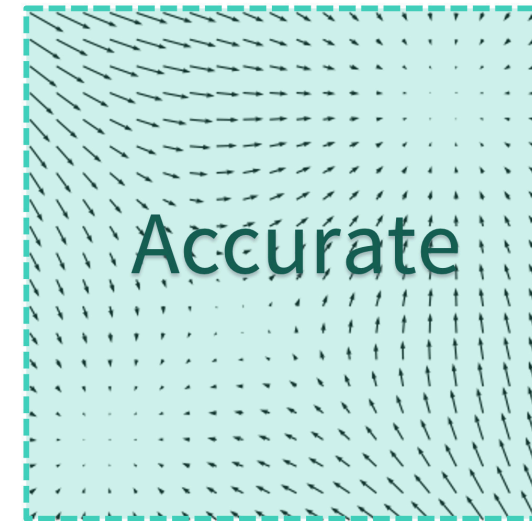
Perturbed density



Perturbed scores



Estimated scores



Noise-Conditional Score-Based Models

As σ_t is **small**,

- (+) Score prediction become more accurate in high-density regions.
- (−) Score prediction become less accurate in low-density regions.

As σ_t is **large**,

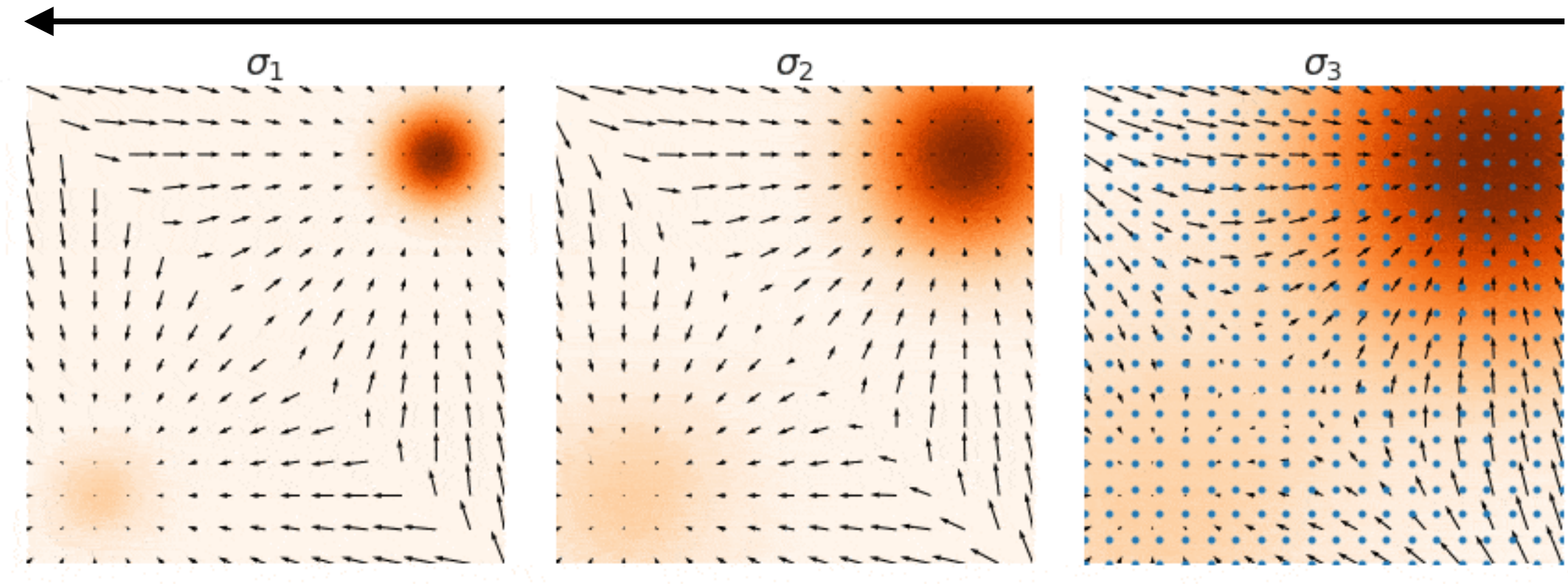
- (+) Score prediction become relatively more accurate in low-density regions.
- (−) Score prediction become relatively less accurate in high-density regions.

Annealed Langevin Dynamics

*Same as the
reverse diffusion process!*

Solution:

Gradually decrease (anneals) σ_t over time!



Annealed Langevin Dynamics

- At the **beginning** (when σ_t is **large**),
we can make good predictions of scores across the entire space, pushing the samples toward the high-density regions of the data distribution.
- As **time progresses** (and σ_t **decreases**),
we achieve better predictions of scores specifically in these high-density regions, allowing for more accurate performance of Langevin dynamics.

Annealed Langevin Dynamics

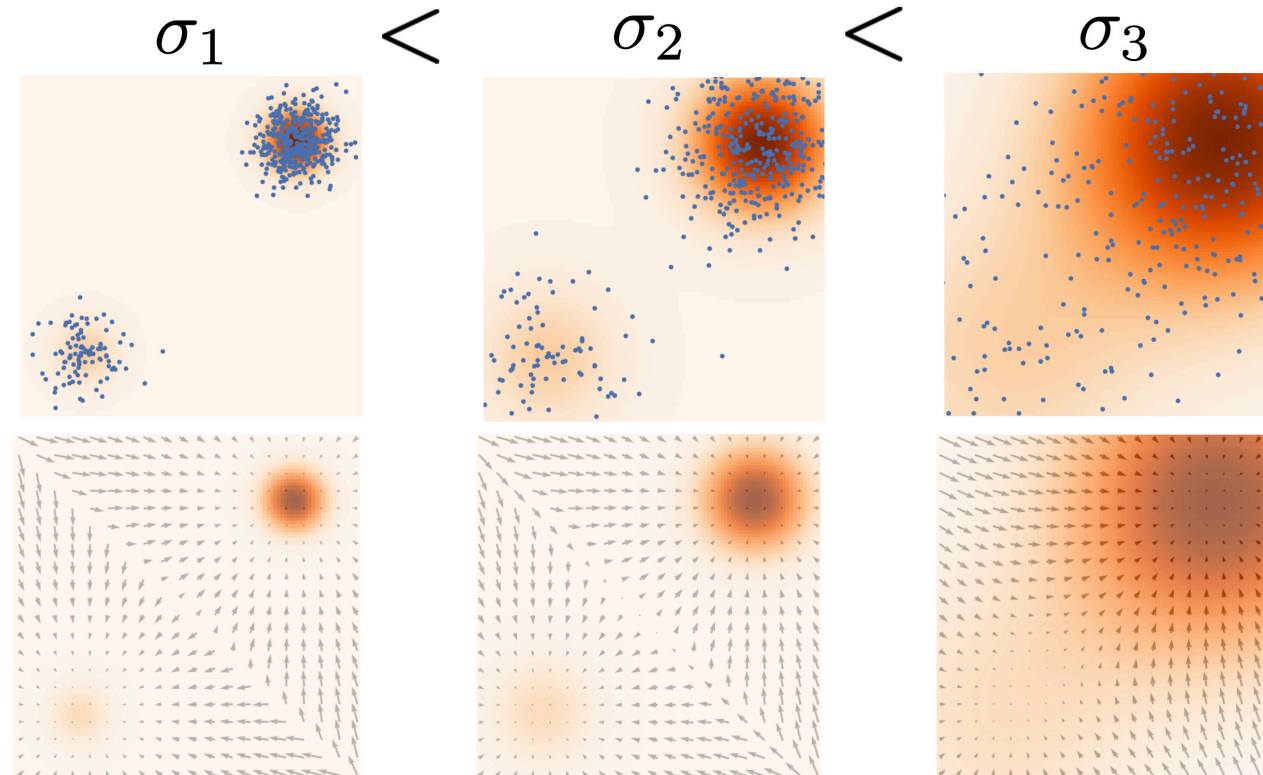
Train the score prediction network $s_\theta(\mathbf{x}_t)$ while **varying** the Gaussian noise σ_t :

$$\mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), t > 1, \mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_0)} \left[\|s_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0)\|^2 \right]$$

Annealed Langevin Dynamics

*Same as the
forward diffusion process!*

The reverse of the annealed Langevin dynamics (the forward process) can be seen as a gradual data perturbation.

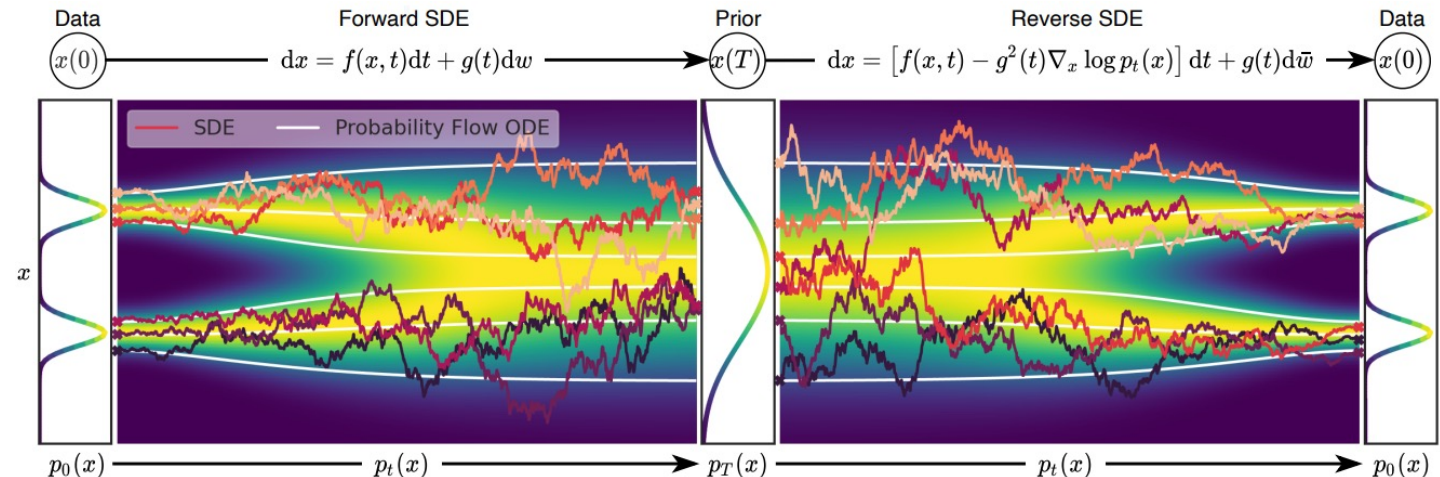


Stochastic Differential Equations

In a **continuous** time domain, the data perturbation (forward) process is described by the following stochastic differential equation (SDE):

$$dx = f(x, t)dt + g(t)d\mathbf{w}$$

- $f(x, t)$: Drift coefficient
- $g(t)$: Diffusion coefficient
- $d\mathbf{w}$: Infinitesimal white noise (called Brownian motion)



Stochastic Differential Equations

Its reverse process is also formulated as another stochastic differential equation:

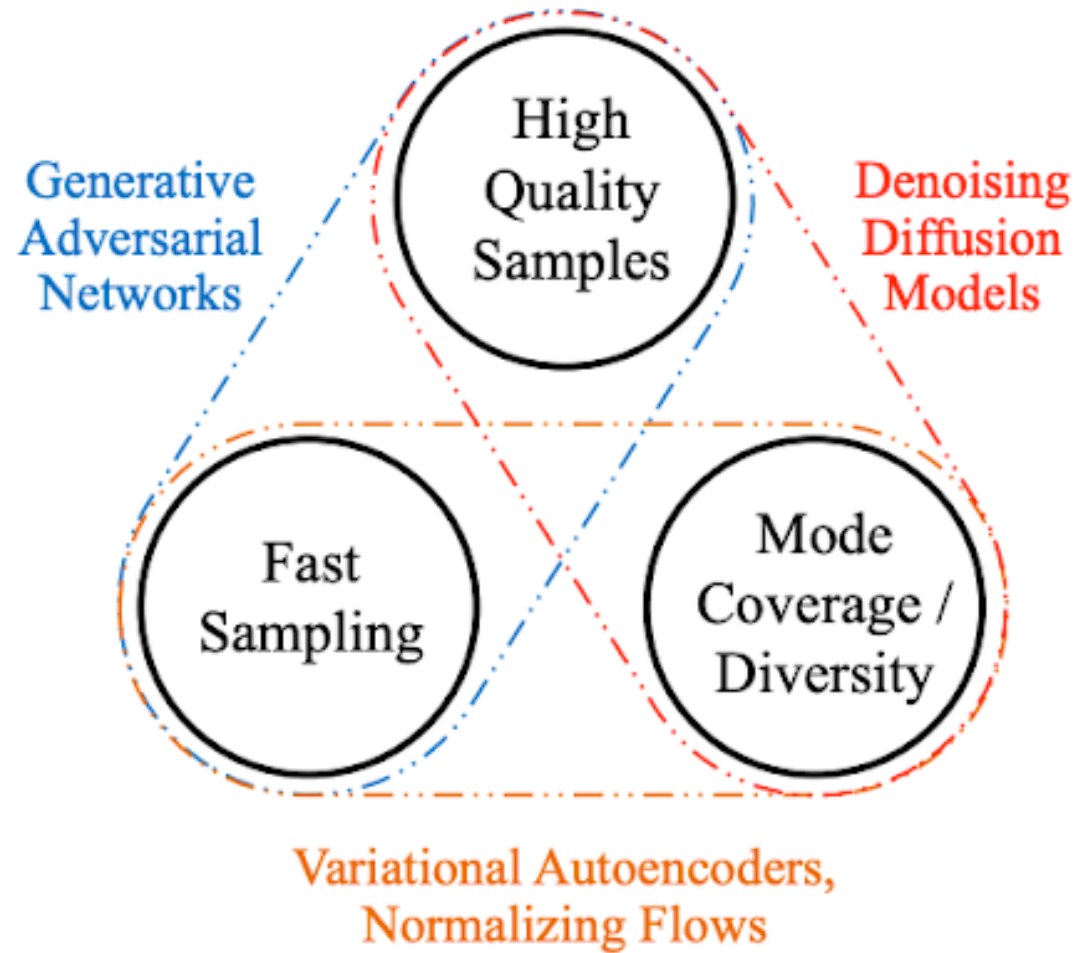
$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t)dt - g^2(t)\nabla_{\mathbf{x}} \log p_t(\mathbf{x})]dt + g(t)d\mathbf{w}$$

DDPM is a specific discretization of the SDE formulations.

Denoising Diffusion Implicit Models (DDIM)

Song et al., Denoising Diffusion Implicit Models, ICLR 2021.

Diffusion Models – Pros and Cons



Diffusion Models – Pros and Cons

- (+) High quality samples
- (+) Diversity
- (–) Very slow speed

How to **speed up** the reverse process while achieving good quality in generation?

Denoising Diffusion Implicit Models (DDIM)

Key features of DDPM :

The sequential forward and reverse processes are **Markovian** processes:

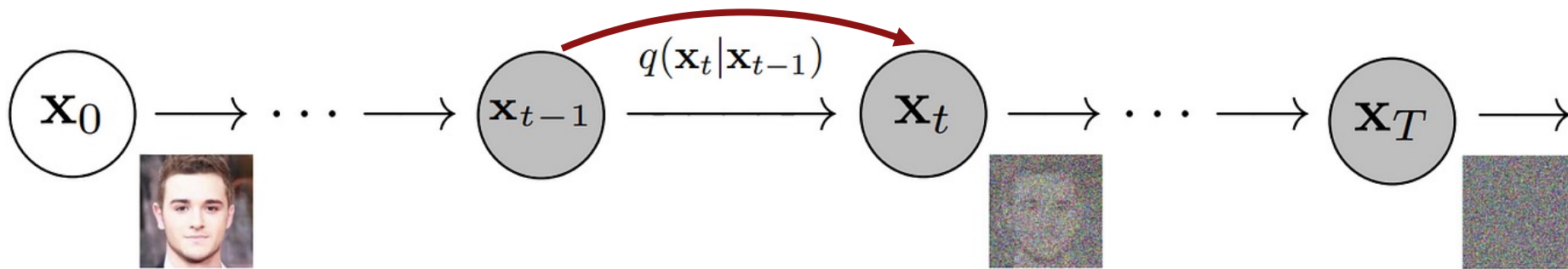
$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0)$$

*Can we consider a **non-Markovian** reverse process?*

Denoising Diffusion Implicit Models (DDIM)

DDPM's Markovian forward process:

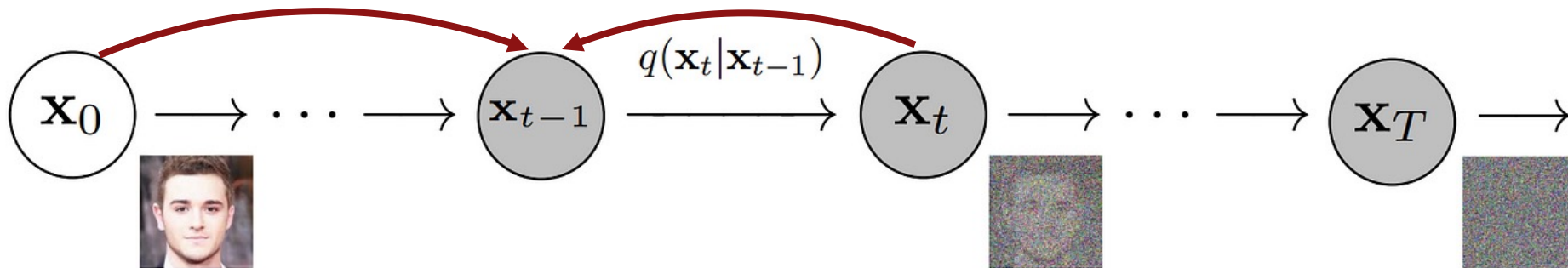
$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$$



Denoising Diffusion Implicit Models (DDIM)

DDIM's non-Markovian forward process:

$$q_{\sigma}(\mathbf{x}_{1:T}|\mathbf{x}_0) = q_{\sigma}(\mathbf{x}_T|\mathbf{x}_0) \prod_{t=2}^T q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$$



Denoising Diffusion Implicit Models (DDIM)

$$q_{\sigma}(\mathbf{x}_{1:T}|\mathbf{x}_0) = \underbrace{q_{\sigma}(\mathbf{x}_T|\mathbf{x}_0)}_{(1)} \prod_{t=2}^T \underbrace{q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}_{(2)}$$

In the forward process,

1. $\mathbf{x}_T$ is sampled from $\mathbf{x}_0$ first.
2. Each $\mathbf{x}_{t-1}$ is sampled from $\mathbf{x}_t$ and $\mathbf{x}_0$ in a **reverse** manner!

Denoising Diffusion Implicit Models (DDIM)

- How to define $q_\sigma(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$?
- Let $q_\sigma(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ be a Gaussian distribution with a **linear mean function** and **variance** σ_t^2 :

$$q_\sigma(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\omega_0 \mathbf{x}_0 + \omega_t \mathbf{x}_t + b, \sigma_t^2 \mathbf{I})$$

- How to determine ω_0 , ω_t , and b ?

Key Idea

How to determine ω_0 , ω_t , and b ?

We want to ensure that $q_\sigma(\mathbf{x}_t|\mathbf{x}_0)$ remains the same as in DDPM:

$$q_\sigma(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$$

Key Idea

Consider an **induction**.

When

$$q_{\sigma}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}),$$

what should ω_0 , ω_t , and b be in order to ensure that

$$q_{\sigma}(\mathbf{x}_{t-1} | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0, (1 - \bar{\alpha}_{t-1}) \mathbf{I})?$$

Key Idea

Hint 1

$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\omega_0 \mathbf{x}_0 + \omega_t \mathbf{x}_t + \mathbf{b}, \sigma_t^2 \mathbf{I})$$

$$q_{\sigma}(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

Marginalization:

$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_0) = \int q_{\sigma}(\mathbf{x}_t|\mathbf{x}_0) q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) d\mathbf{x}_t$$

Key Idea

Hint 2

When $p(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \sigma_x^2 \mathbf{I})$ and $p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(a\mathbf{x} + b, \sigma_y^2 \mathbf{I})$,

$$\begin{aligned} p(\mathbf{y}) &= \int p(\mathbf{x})p(\mathbf{y}|\mathbf{x}) d\mathbf{x} \\ &= \mathcal{N}(a\boldsymbol{\mu} + b, (\sigma_y^2 + a^2 \sigma_x^2) \mathbf{I}) \end{aligned}$$

Key Idea

Q1. What are the mean and variance $q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_0)$ with respect to ω_0 , ω_t , and b ?

Q2. What are ω_0 , ω_t , and b ?

Key Idea

A1. $q_\sigma(\mathbf{x}_{t-1}|\mathbf{x}_0) =$

$$\mathcal{N}\left(\omega_0\mathbf{x}_0 + \omega_t\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0\right) + b, \left(\sigma_t^2 + \omega_t^2(1 - \bar{\alpha}_t)\right)\mathbf{I}\right)$$

$$\mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0, (1 - \bar{\alpha}_{t-1})\mathbf{I}\right)$$

Key Idea

$$\mathbf{A2.} \quad \omega_t = \sqrt{\frac{1 - \bar{\alpha}_{t-1} - \sigma_t^2}{1 - \bar{\alpha}_t}}$$

$$\omega_0 = \sqrt{\bar{\alpha}_{t-1}} - \sqrt{\bar{\alpha}_t} \sqrt{\frac{1 - \bar{\alpha}_{t-1} - \sigma_t^2}{1 - \bar{\alpha}_t}}$$

$$b = 0$$

$$\therefore q_\sigma(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\omega_0 \mathbf{x}_0 + \omega_t \mathbf{x}_t + b, \sigma_t^2 \mathbf{I})$$

$$= \mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2 \mathbf{I}\right)$$

Denoising Diffusion Implicit Models (DDIM)

$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2 \mathbf{I}\right)$$

with arbitrary σ_t^2 guarantees that $q_{\sigma}(\mathbf{x}_t|\mathbf{x}_0)$ remains the same as in DDPM!

DDPM vs. DDIM

DDPM

- $q(\mathbf{x}_t | \mathbf{x}_{t-1})$ is defined.
- $q(\mathbf{x}_t | \mathbf{x}_0)$ and $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$ are derived from $q(\mathbf{x}_t | \mathbf{x}_{t-1})$.

DDIM

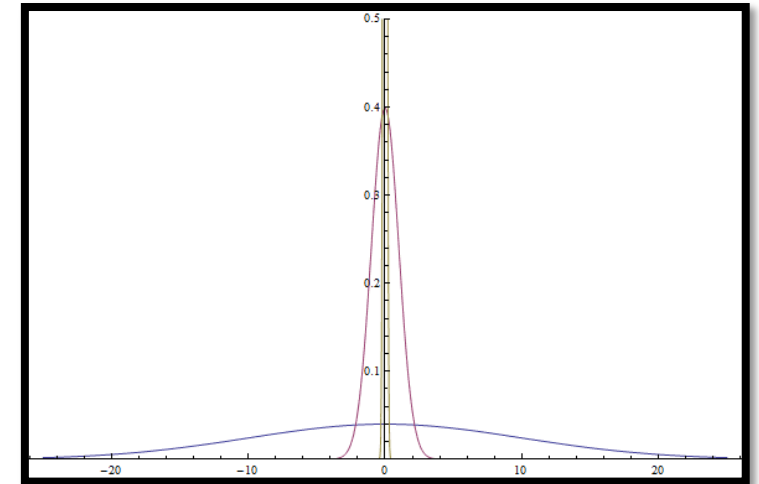
- $q_\sigma(\mathbf{x}_T | \mathbf{x}_0)$ and $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$ are defined.

Denoising Diffusion Implicit Models (DDIM)

$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2 \mathbf{I}\right)$$

[Important] What if we set $\sigma_t^2 = 0$?

Then the forward and reverse processes become **deterministic**!



Loss Function

- The **ELBO remains unchanged** from DDPM, up to a constant.
- The variational likelihood distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$ in the reverse process is learned by minimizing the **same loss function** as in DDPM:

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

Loss Function

[Important] No need to retrain the noise predictor!

- The noise predictor $\hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$ trained for **DDPM** can be directly used in the **DDIM** reverse process!
- Using the same noise predictor $\hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$, you can choose to perform either the DDPM or the DDIM reverse process.

DDPM Reverse Process

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0, \left(\frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t\right) \mathbf{I}\right)$$

For each time step $t = T, \dots, 1$, repeat:

1. Compute $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t))$
2. Compute $\tilde{\boldsymbol{\mu}} = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_{0|t}$.
3. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. Compute $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \sqrt{\frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}} \beta_t \mathbf{z}_t$.

DDIM Reverse Process

$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2 \mathbf{I}\right)$$

For each time step $t = T, \dots, 1$, repeat:

1. Compute $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}}(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t))$
2. Compute $\tilde{\boldsymbol{\mu}} = \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_{0|t}}{\sqrt{1 - \bar{\alpha}_t}}.$
3. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. Compute $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \sigma_t \mathbf{z}_t.$

Denoising Diffusion Implicit Models (DDIM)

What is the meaning of DDIM?

- It's a generalization of DDPM!
- DDPM is a special case of DDIM when

$$\sigma_t^2 = \tilde{\sigma}_t^2 = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t.$$

- In this case, DDIM operates as a Markovian process.

Controlling Stochasticity

Let's parameterize σ_t as $\sigma_t = \eta \sqrt{\frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}} \beta_t$.

- $\eta = 0$: **Deterministic** process.

For the same $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, we always obtain the same $\mathbf{x}_0$.

- $\eta = 1$: Same as **DDPM**.

Back to the Pros and Cons...

- (+) High quality samples
- (+) Diversity
- (−) Very slow speed

Accelerating Sampling Process

The DDPM/DDIM reverse process with the full sequence of time steps $t \in [1, 2, \dots, T]$:

$$p_{\theta}(\mathbf{x}_{0:T}) = p_{\theta}(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t)$$

Accelerating Sampling Process

Consider a **sub-sequence** of the time steps:

$$\tau = [\tau_1, \tau_2, \dots, \tau_S].$$

The reverse process for this **sub-sequence**

$$p_{\theta}(\mathbf{x}_{\tau}) = p(\mathbf{x}_T) \prod_{t=1}^S p_{\theta}(\mathbf{x}_{\tau_{i-1}} | \mathbf{x}_{\tau_i})$$

is optimized using the same objective function as in the full sequence.

Accelerating Sampling Process

- As **smaller** time steps are used, the quality of the generated data can worsen.
- However, quality degradation is **mitigated** when the DDIM reverse process becomes more **deterministic**.

Accelerating Sampling Process

FID scores (lower is better) while varying

- η : Stochasticity
- S : The number of time steps

S	CIFAR10 (32×32)					CelebA (64×64)					
	10	20	50	100	1000	10	20	50	100	1000	
η	0.0	13.36	6.84	4.67	4.16	4.04	17.33	13.73	9.17	6.53	3.51
	0.2	14.04	7.11	4.77	4.25	4.09	17.66	14.11	9.51	6.79	3.64
	0.5	16.66	8.35	5.25	4.46	4.29	19.86	16.06	11.01	8.09	4.28
	1.0	41.07	18.36	8.01	5.78	4.73	33.12	26.03	18.48	13.93	5.98

Accelerating Sampling Process

When $\eta = 1$ (DDPM), the quality gets worse quickly as S decreases from 1000 to 10.

	S	CIFAR10 (32×32)					CelebA (64×64)				
		10	20	50	100	1000	10	20	50	100	1000
η	0.0	13.36	6.84	4.67	4.16	4.04	17.33	13.73	9.17	6.53	3.51
	0.2	14.04	7.11	4.77	4.25	4.09	17.66	14.11	9.51	6.79	3.64
	0.5	16.66	8.35	5.25	4.46	4.29	19.86	16.06	11.01	8.09	4.28
	1.0	41.07	18.36	8.01	5.78	4.73	33.12	26.03	18.48	13.93	5.98

Accelerating Sampling Process

- When $\eta = 0$ (deterministic),
- the quality does not get bad too much even when $S = 10$.
- The quality when $S = 1000$ is even better than DDPM!

S	CIFAR10 (32×32)					CelebA (64×64)					
	10	20	50	100	1000	10	20	50	100	1000	
η	0.0	13.36	6.84	4.67	4.16	4.04	17.33	13.73	9.17	6.53	3.51
	0.2	14.04	7.11	4.77	4.25	4.09	17.66	14.11	9.51	6.79	3.64
	0.5	16.66	8.35	5.25	4.46	4.29	19.86	16.06	11.01	8.09	4.28
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